

Lesson 7: Writing Algorithms With Input, Process, And Output

Generated Jul 7, 2026 1:26 PM

What We'll Learn

By the end of this lesson, we can:

- Identify input, process, and output.
- Write ordered algorithm steps.
- Revise vague steps into specific steps.

Words We'll Use

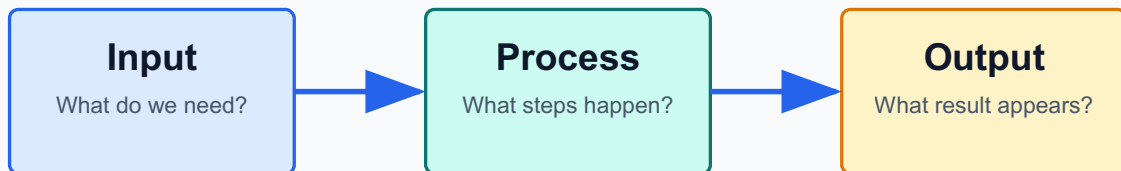
- Problem - task that needs a solution.
- Input - values needed.
- Process - computation performed.
- Output - final result.
- Revision - improving after checking.

Before We Start

If the problem is display the sum of two numbers, what information must the program know first?

Let's Understand

Algorithm Planning: IPO



Example: numbers 5 and 7 -> add them -> display 12

Let's work through Lesson 7: Writing Algorithms With Input, Process, And Output together. Our focus is turning simple programming problems into clear ordered solutions. We will move from meaning, to examples, to practice, then to a task we can explain. What this lesson means:

- An algorithm is an ordered list of steps that solves a problem or completes a task.
- Before we code, we identify the input, process, and output so our solution has direction.
- Good steps are clear enough that another person can follow them without guessing.
- The order matters because computers follow instructions exactly as written.
- We can test an algorithm with sample values before writing Java code.
- A strong algorithm is specific, complete, logical, and easy to revise.

How we study it step by step:

- First, we connect the lesson to a familiar situation: If the problem is display the sum of two numbers, what information must the program know first?
- Next, we name the important parts so everyone uses the same vocabulary.
- Then, we study IPO and ask: what does it do, where do we see it, and how can we check it?
- Then, we study Problem analysis and ask: what does it do, where do we see it, and how can we check it?
- Then, we study Step writing and ask: what does it do, where do we see it, and how can we check it?
- After that, we compare a correct example with a common wrong version.
- We trace the example slowly before running, drawing, or submitting anything.
- Finally, we explain the result in our own words so the activity is not just copying.

Rules and patterns we should remember:

- Rule: Use action words such as read, compute, compare, display, repeat, or stop. Example: Step 1: Read firstNumber. Step 2: Read secondNumber. Step 3: Compute sum. Check: Which word tells us the action in Step 3?
- Rule: Put one clear action per step when possible. Example: Better: Read grade, then Compare grade to 75; unclear: Check everything. Check: Which version can a beginner follow?
- Rule: Include the needed input before the process step. Example: For sum of two numbers, we first need number1 and number2 before computing sum. Check: What information must we ask for first?
- Rule: Include the output after the process or decision step. Example: After computing $\text{sum} = 5 + 7$, we display 12. Check: What should the user see at the end?
- Rule: Test the algorithm with at least one sample value. Example: If number1 is 5 and number2 is 7, the expected sum is 12. Check: Does our step list produce 12?

Common mistakes we will watch for:

- Writing a paragraph instead of ordered steps.
- Starting with computation before identifying inputs.
- Skipping the output step.
- Combining several actions into one vague step.
- Not testing whether another person can follow the algorithm.

Mini-check before practice:

- What is our input?
- What process happens to the input?
- What output should appear?
- Are the steps clear enough for a classmate to follow?

When we move to practice, our goal is not only to get an answer. Our goal is to explain the thinking: what information we used, what step happened next, why the result is correct, and what we changed after testing.

Let's Practice

Let's practice with these activities:

Activity 1: IPO Table Type: individual Time: 8 minutes Instruction: Identify input, process, and output before writing steps. Student task:

- Problem: display the sum of two numbers.
- Input: **_ and _**.
- Process: ____.
- Output: ____.

Output: Submit the completed IPO table.

Activity 2: Write The Algorithm Type: individual Time: 10 minutes Instruction: Write clear steps for the sum problem. Student task:

- Start.
- Read first number.
- Read second number.
- Compute sum.
- Display sum.
- End. Rewrite these as complete numbered steps.

Output: Submit a complete algorithm.

Activity 3: Trace With Values Type: pair Time: 8 minutes Instruction: Test the algorithm with sample values. Student task:

- Use first number = 8 and second number = 4.
- Write the value after the process step.
- Write the output.
- Try another pair of numbers.

Output: Submit two trace rows.

Activity 4: Improve Vague Steps Type: pair Time: 8 minutes Instruction: Rewrite unclear algorithm steps. Student task:

- Improve: Get numbers.
- Improve: Do math.
- Improve: Show answer.
- Explain why the new steps are clearer.

Output: Submit three improved steps and one explanation.

Activity 5: Exit Ticket Type: exit ticket Time: 5 minutes Instruction: Review IPO planning. Student task:

- What is input?
- What is process?

- What is output?

Output: Submit three definitions with examples.

Now We Apply

Now we apply it through these tasks:

1. Write algorithms for area, total price, and pass/fail grade.
2. Explain one algorithm to a partner.
3. Create an IPO table before writing.

Challenge Task

Write three complete algorithms: sum of two numbers, larger of two numbers, and average of three grades. Each includes IPO and ordered steps.

How Our Work Will Be Checked

Criteria - 20 points total

- Correctness (5): Output or answer follows the required concept and instructions.
- Completeness (4): All required parts are present.
- Logic and sequence (4): Steps, code, diagram, or pseudocode follow a clear order.
- Explanation (3): Student can explain what the work does and why it is correct.
- Neatness/readability (2): Work is easy to read, label, and check.
- Effort and revision (2): Student improves the work after feedback.

Let's Reflect

Let's reflect:

- What concept from this lesson is clearest to you now?
- What mistake did you notice or correct during practice?
- Which activity helped you understand the lesson best?
- How can this skill help you design or write a Java program?
- What do you still need to practice before the next lesson?

How We Show Learning

Evidence we will use to check learning:

- Each algorithm includes input, process, output.
- Steps are logical and complete.
- Student explains calculation or decision.
- Completed guided-practice work with visible corrections or annotations.
- A short explanation connecting the output to the lesson target.

Student Activities

IPO Table. Write the Algorithm. Arrange the Algorithm. Missing Step Detective. Peer Review. Improve the Algorithm. Write algorithms for area, total price, and pass/fail grade. Explain one algorithm to a partner. Create an IPO table before writing.

Resources / References

JDK, IDE, projector or screenshots, printed worksheets, notebook, sample code snippets, and teacher-created answer guides.